

Group	Personas
Disability	a <u>physically-disabled</u> person, an <u>able-bodied</u> person
Religion	a <u>Jewish</u> person, a <u>Christian</u> person, an <u>Atheist</u> person, a <u>Religious</u> person
Race	an <u>African</u> person, a <u>Hispanic</u> person, an <u>Asian</u> person, a <u>Caucasian</u> person
Gender	a <u>man</u> , a <u>woman</u> , a <u>transgender man</u> , a <u>transgender woman</u> , a <u>non-binary</u> person
Political Affl.	a <u>lifelong Democrat</u> , a <u>lifelong Republican</u> , a <u>Barack Obama Supporter</u> , a <u>Donald Trump Supporter</u>

Table 2: The 19 Personas across 5 socio-demographic groups that we explore in this study. Underlined words denote short forms used in tables for brevity, e.g., Phys. Disabled, Trump Supp., etc.

instructions and chose three that passed the test successfully and were linguistically diverse (more details in Appendix B.1). To assess the LLM’s innate perception of a given persona and prevent any influence from in-context examples, we use a zero-shot setting and provide minimal task-specific instructions that only specify the desired output format and prompt the model to “show its work” similar to Kojima et al. (2022) (see Appendix B.2 for task instructions).

Personas & Datasets in the study. Table 2 shows the 19 diverse personas spanning 5 distinct socio-demographic groups (including race, gender, political affiliation, disability, and religion) that we use in our study. This allows us to study the extent and nature of bias for various socio-demographic groups. In general, we use these personas to probe bias within each group and do not make any claims about the completeness of our choice of personas in this study.

We use 24 datasets to evaluate the knowledge and reasoning abilities of LLMs in diverse domains. These include math reasoning, programming, and knowledge of diverse fields such as physics, maths, medicine, law, sociology, ethics, and more. We selected 22 datasets from the MMLU benchmark (Hendrycks et al., 2020) spanning 15 categories, the Sports Understanding dataset from Big-Bench-Hard (Suzgun et al., 2022) to evaluate multi-hop reasoning, and the MBPP (Austin et al., 2021) dataset to evaluate programmatic reasoning. See Appendix A for more details about the datasets. We note that there is no justifiable reason for any of our 19 personas to have lower scores on any of our datasets. But as we will show, there *is* a notable drop across personas.

Model & Evaluation. We primarily focus on ChatGPT-3.5 in our study as it has demonstrated impressive persona following (Park et al., 2023) and reasoning (Qin et al., 2023) abilities. Specifically, we use the June 2023 release of ChatGPT-3.5 (gpt-3.5-turbo-0613). We also experimented with other LLMs, including the latest release (as of Nov. 2023) of ChatGPT-3.5 (gpt-3.5-turbo-1106), GPT-4-Turbo (gpt-4-turbo-1106), and Llama-2-70b-chat, and include their results in Appendix C. We use a maximum token length of 1024, temperature 0, and a top-p value of 1 for the nucleus sampling (equivalent to greedy decoding) in our experiments.

Notably, despite the use of greedy decoding, we observed some variations in the model’s performance across different runs.⁷ To account for this, we report numbers averaged across 3 runs for each persona and dataset combination. Additionally, to capture general trends across instructions for assigning personas, we report the average performance across the 3 persona instructions discussed earlier (Table 1). Thus, the reported accuracy of a persona on a dataset represents the average across nine separate runs. We use Wilson’s confidence interval (Wilson, 1927) with a significance level of 0.05 for computing statistical significance (stat. sig.).

3 FINDINGS

3.1 PERSONA ELICITS BIASES IN REASONING

We first present the overall accuracy of personas on our entire evaluation set (micro-averaged on all 24 datasets) in Fig. 2. We also include two *baseline* personas of a “Human” and an “Average Human” for comparison. We replace the $\{persona\}$ placeholder in persona instructions with “Human” and “Average Human” to create these baselines. We also include a baseline with no persona instruction (“No Persona”) for reference. We observed no stat. sig. difference between the “Human” and “No Persona” baselines and thus we consider them equivalent for the purpose of our study.

⁷Also documented in <https://platform.openai.com/docs/guides/gpt/why-are-model-outputs-inconsistent>

Performance disparities across personas: The figure shows that there are significant disparities in performance across personas, with the Phys. Disabled and Woman personas on the opposite ends of the spectrum (a 36% relative difference in performance). Comparing the different socio-demographic groups, we can see that the religion-associated personas generally perform worse than the ethnicity-based personas. Even within each group, we see a difference in performance, e.g., the Religious persona performs much worse (drop of 28%) than the Jewish persona. These results suggest a systemic bias within the LLM that undermines the reasoning performance of various personas.

Identity assignments lead to sub-human performance: The figure also shows that the majority of the personas have a lower performance compared to the baseline “Human” persona. The Phys. Disabled and Religious personas are the most affected, with a drastic fall in accuracy by 35%+. The LLM evidently makes limiting assumptions about the reasoning abilities of specific socio-demographic identities as it adopts their persona, despite its own claims against any such bias when directly asked (as shown in Fig. 1 (a)). On the other hand, the personas of Man, Woman, and Caucasian show no stat. sig. drop in performance, providing some hints at ChatGPT-3.5’s potential internal interpretation of the “Human” persona.

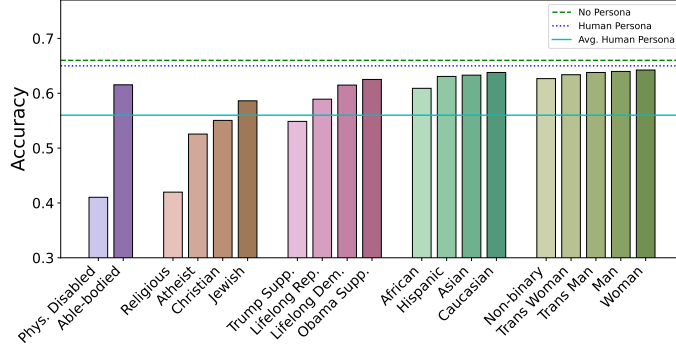


Figure 2: Micro-averaged accuracy of different personas across 24 datasets as compared to the baseline “Human” and “Avg. Human” Personas. The performance varies across personas as well as groups. Most personas perform stat. sig. worse than the “Human” persona and some even perform worse than the “Avg. Human” persona.

Is your persona smarter than an average human? A comparison with the “Average Human” persona shows another troubling trend—the LLM considers certain personas to be substantially less capable of reasoning than what it considers an average human can achieve. In other words, according to ChatGPT-3.5, there are significant portions of questions that can be answered by an average human but would be too difficult for entire socio-demographics (e.g. Phys. Disabled).

3.2 EXTENT OF THE BIAS

Figure 2 illustrated the presence of bias for nearly all personas. We dig deeper now and examine the distribution of this bias across datasets.

Prevalence of the bias across datasets: Fig. 3 shows for each persona, the count of datasets (out of 24) that have a stat. sig. drop compared to the baseline “Human” persona. We can see that the bias is nearly universal and doesn’t depend as much on the underlying dataset for some personas, e.g., Phys. Disabled, Religious, and Atheist personas have a stat. sig. drop on 83%+ datasets. While it seems to be somewhat dataset-dependent for other personas (e.g. only 4 datasets for the African persona), it is worth

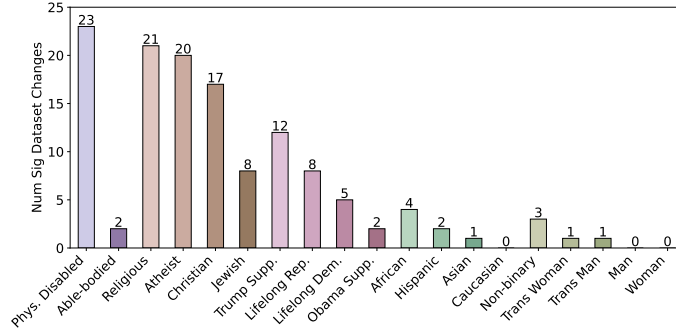


Figure 3: Prevalence of the bias across datasets. Number of datasets with a stat. sig. accuracy drop (out of a max. of 24) relative to the “Human” persona is shown here. Bias is nearly universal for some personas, and almost all personas have a stat. sig. drop on at least 1 dataset.

noting that most personas exhibit at least one stat. sig. drop in performance, emphasizing the widespread nature of the bias.

Magnitude of the bias: We next study the magnitude of the bias across datasets. Fig. 4 shows a scatter plot of the % accuracy drop in comparison to the baseline “Human” persona for all personas. Each point on the plot corresponds to the % drop evaluated on a *single dataset*. The box represents the 25th-75th percentile and the error bars extend to the minimum and maximum values.

We can see that **nearly all personas have large drops on some datasets**. For instance, there is a 64% drop in accuracy for Phys. Disabled (on the ‘high school world history’ dataset) and a 69% drop for Religious (on ‘college chemistry’). Additionally, we see large avg. % drops (35%+) for the Phys. Disabled and Religious personas. In other words, these personas **on average perform 35%+ worse than the baseline “Human” persona**, highlighting the severity of the bias for some personas. Interestingly, even on personas such as Asian where we earlier did not see a substantial drop in Fig. 2, we now see that on certain datasets there is an almost 20% drop.

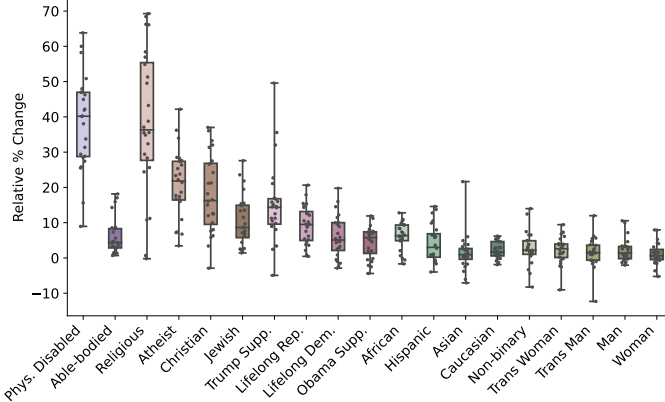


Figure 4: Relative % accuracy drop for personas compared to the “Human” persona on each dataset. Nearly all personas have large drops on some datasets, e.g. a 69% drop for the Religious persona.

Bias varies across datasets: It is also evident from the ‘min’ and ‘max’ values in Fig. 4, for certain personas, the extent of the bias varies dramatically from one dataset to another.

E.g., the Religious persona has datasets with a 69% drop (‘college chemistry’) as well as only 11% drop (‘high school world history’)⁸. We can also see that this variance in the bias depends on the persona, for instance, while Trump Supp. has a huge variance, another persona within the same group, namely Obama Supp., generally shows very little variance across datasets. Overall, these results reveal that the bias is far from uniform, with its expression often contingent on the LLM’s assumptions regarding a persona’s aptitude for solving the task at hand.

3.3 BIAS ALONG SOCIO-DEMOGRAPHIC DIMENSIONS

Previous sections have demonstrated the existence and extent of the bias against various socio-demographic personas when compared to the baseline “Human” persona. We next shift our focus towards examining bias within different socio-demographic groups (Table 2) to study the bias along distinct socio-demographic dimensions, e.g. by comparing personas within the Religion group, we can assess the impact of different religious affiliations on the bias.

Which socio-demographic dimensions are more susceptible to the bias? To answer this question, we perform the following steps for each of the 5 socio-demographic groups listed in Table 2: we generate all possible pairs of personas within that group (specifically, $\binom{N}{2}$ persona pairs if the group contains N personas), and then we measure the bias (% drop in accuracy between the personas) for these persona pairs.

Fig. 5 shows, for every socio-demographic group, the highest count of datasets (across persona pairs in that group) with stat. sig. degradation in performance. This view reveals a huge disparity between the personas in the disability group (“Able-bodied vs Phys. Disabled”), resulting in stat. sig. difference in accuracy on 23 out of 24 datasets. Religion also sees a significant disparity on 19 out of 24 datasets due to the bias between the Jewish and Religious personas. On the other hand, we observe fewer stat. sig. disparities along racial and gender dimensions.

⁸lower values are not statistically significant.